

Initial System Characterization

Target water output capacity: 30 L/day

Number of heat exchangers: 2

Preliminary operating conditions

Parameter	Value	Description
T_{amb}	25 °C	Ambient temperature of adsorption air
RH_{amb}	50-70 %	Ambient relative humidity of adsorption air (inlet)

Preliminary Constraints

Parameter	Value	Description
T_{max}	70 °C	Maximum sorbent temperature (not a hard constraint, but the hard constraint is temperature at which water could boil, 100 °C, or at which there is uncontrolled pressure generation)
Δp_{max}	40 Pa	Maximum pressure drop across heat exchangers
τ_c	8-10 hours	Cycle duration (for full cycle of from adsorption to reset)

Key Performance Metrics

Parameter	Equation	Description
SEC	$SEC = \frac{Q_{in}}{V_{water}}$	Specific energy consumption: ratio of thermal energy supplied to actual volume of water produced. Aim to minimize.
$\eta_{thermal}$	$\eta_{thermal} = \frac{m_w h_{fg}}{Q_{in}}$	Idealistic thermodynamic efficiency of the system. Aim to maximize.
SLP	$SLP = \frac{V_{device}}{V_{sorb}}$	Sorbent layer packing: quantifies the use of device volume with respect to the minimum size of an ideal device (which would only contain the sorbent bed). Measures efficient use of space. Aim to minimize.

η_{AWH}	$\eta_{AWH} = \frac{\left(V_{\text{water}}^{\text{day}} \right)^{\text{actual}}}{\left(\frac{m_{\text{sorb}} \text{ uptake Swing } N_{\text{cycles/day}}}{\rho_{\text{water}}} \right)}$	Water harvesting efficiency: Weighs the actual water production of system with the theoretical maximum output of the sorbent for a given environmental condition. Aim to maximize.
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The purpose of this project is a research unit, aimed to automate the manual process of AWH to produce a large amount of water per day (30 L/day), while being cheaper and more energy efficient than other automated AWH methods. Hence, the following optimization goals become the most important for the purposes of this project:

1. Maximize η_{AWH}
2. Maximize η_{thermal}

SLP is less important, since production and manufacturing for scale is not of primary concern here.

Fixed Sorbent Material Properties

Parameter	Value	Equation	Description
h_s	*Need from Joseph*		Adsorption enthalpy. Determines how much energy to be supplied to the sorbent for desorption
$Uptake$	*Need from Joseph*	$f(RH, T)$	Describes the water uptake of the sorbent (in g of water per g of sorbent) as a function of local RH at a constant temperature for the given sorbent. Described by sorption isotherm.
D_v	*Need from Joseph*		Intercrystalline diffusivity: Affects sorption kinetics
D_μ	*Need from Joseph*		Intracrystalline diffusivity: Affects sorption kinetics
ρ_{sorb}	*Need from Joseph*		Sorbent density
m_{sorb}	*Need from Joseph*		Total sorbent mass used

With this, for a given $RH_{\text{amb}} \approx 60\%$ at $T_{\text{amb}} = 25^\circ\text{C}$, we can use the sorption isotherm for the material to determine the theoretical adsorption uptake capacity of the sorbent.

Physical Design Parameters

Parameter	Value	Equation	Description
L	1 m (constant)	-	Length of sorbent bed (heat exchanger side length)
t_s	*Need from Joseph* (constant)	-	Sorbent layer thickness on sorbent bed (heat exchanger)
t_g		-	Air gap between sorbent layers on heat exchangers
ϵ	0.67 (assumption)	$\epsilon = 1 - \frac{\rho_b}{\rho_{sorb}}$	Porosity of sorbent bed. Highly dependent on the packing of the sorbent on the scaffolding material, and usually 0.67 is used when the bed is fixed.
μ_{avg}	3-5 m/s (preliminary estimation)	-	Average airflow speed over heat exchangers
$N_{cycles/day}$	3	$\frac{24 \frac{hrs}{day}}{\tau_c}$	Number of complete cycles per day
τ_{ads}	4 hours (preliminary estimation)	$\frac{\tau_c}{2}$	Cycle time allocated for adsorption. Starting estimate is an even split between adsorption and desorption stages.
τ_{des}	4 hours (preliminary estimation)	$\frac{\tau_c}{2}$	Cycle time allocated for desorption + condensation + reset. Starting estimate is an even split between adsorption and desorption stages.
α	Aim to maximize	$\alpha(t) = \frac{RH_{in} - RH_{out}(t)}{RH_{in}}$	Vapor recovery ratio: percentage of water vapor that is adsorbed by the sorbent from the supplied airflow
$Uptake_{\tau}$	Aim to maximize	Empirically determined	Uptake of the sorbent after a fixed time (paper describes time to be $\tau = 30$ minutes). For purposes of project, $\tau = \tau_{ads}$)
T_{desorp}	65 °C (preliminary estimate)		Desorption temperature

			(temperature to which sorbent is to be heated to release water, and controlled at)
F_{wg}	2-4 L/min		Water-glycol flow rate through heat exchanger. Must be sufficient to maintain effective temperature difference across heat exchanger

Vapor recovery ratio (α) and the uptake within the specified adsorption cycle time are key device-level performance parameters that are affected by t_s , t_g , ϵ , μ_{avg} . Since the sorbent material is provided by Joseph, t_s and ϵ are assumed to be fixed. Hence, the air gap between sorbent-coated heat exchangers (t_g) and average airflow speed (μ_{avg}) become key adjustable parameters for design.

In the paper, simulations were conducted to determine the effect of changing t_s , t_g , ϵ , μ_{avg} on vapor recovery ratio and uptake (using a linear driving force model) for a specific sorbent and sorbent bed. Similar simulations will need to be conducted to determine the effect of varying our variables of interest (t_g and μ_{avg}) on vapor recovery ratio and uptake, to empirically determine the system behaviour upon changing those parameters, if any at all.

Fan power can be determined using the following equations from the parameters above, to guide the selection of fans:

Heat used as input can be determined using the following equation: